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2027 CYCLE

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CUET UG

Chemistry

Electrochemistry

Official Syllabus Unit II · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit II)	Skills the questions test
Galvanic cells and electrode potential	Electrochemical Cells, Galvanic Cells, Standard Electrode potential and its measurement, EMF of cell	Labelling a Daniell cell, writing cell notation, computing E°_{cell} from standard electrode potentials, and using the SHE as the zero reference.
Nernst equation and Gibbs energy	Nernst Equation & its applications (calculation of equilibrium constant and E cell calculations), Electrochemical Cell & Gibbs Energy of the reaction	Applying $E = E^\circ - \left(\frac{0.059}{n}\right)\log Q$, converting E° to K and to $\Delta G^\circ = -nFE^\circ$, and reading what each sign means.
Conductance and Kohlrausch law	Conductance of electrolytic Solutions, Measurement of the conductivity of Ionic Solutions, Specific & Molar Conductivity, Variation of Conductivity and molar conductivity with Concentration, Kohlrausch law & its applications	Distinguishing κ from Λ_m, predicting how each behaves on dilution, and computing α and K_a from molar conductivities.
Electrolysis and Faraday's laws	Electrolytic Cells & Electrolysis, Faraday's Laws of Electrolysis, Products of Electrolysis	Predicting anode/cathode products in molten vs aqueous salts and computing deposited masses with $w = ZIt$.
Batteries, fuel cells and corrosion	Batteries- Primary & Secondary batteries, fuel cell, Corrosion	Classifying dry cell, mercury cell, lead-acid and Ni-Cd batteries, describing the $H_2 - O_2$ fuel cell, and naming corrosion-prevention methods.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with

+5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit II turns redox chemistry into numbers: cell voltages, the Nernst equation, conductivities and Faraday's laws.

Galvanic Cells: Chemistry That Pushes Electrons

Drop a strip of zinc into blue copper sulphate solution and watch the blue fade while a red-brown coat grows on the metal. Zinc atoms are handing electrons to copper ions:

$$\mathrm{Zn}(s) + \mathrm{Cu}^{2+}(\mathrm{aq}) \rightarrow \mathrm{Zn}^{2+}(\mathrm{aq}) + \mathrm{Cu}(s)$$

. In the beaker, that electron hand-over happens on the spot and the energy leaks away as useless heat. **Electrochemistry** is the art of separating the two halves of that hand-over so the electrons are forced to travel through a wire first — and a flowing wire-full of electrons is exactly what we call an electric current. A device that does this from a **spontaneous** reaction is a **galvanic cell** (also called a voltaic cell), and the syllabus's model example is the Daniell cell.

- **Two half-cells:** the Daniell cell splits the action. A zinc rod stands in

  $$\mathrm{ZnSO_4}$$

  solution in one beaker; a copper rod stands in

  $$\mathrm{CuSO_4}$$

  solution in the other.
- **Oxidation at the anode:** the zinc rod loses electrons —

  $$\mathrm{Zn}(s) \rightarrow \mathrm{Zn}^{2+}(\mathrm{aq}) + 2e^-$$

  — so the rod slowly dissolves. In a galvanic cell the anode is the **negative** terminal.
- **Reduction at the cathode:** copper ions collect those electrons —

  $$\mathrm{Cu}^{2+}(\mathrm{aq}) + 2e^- \rightarrow \mathrm{Cu}(s)$$

  — and the rod gains mass. The cathode is the **positive** terminal.
- **The salt bridge:** an inverted U-tube of agar jelly soaked in KCl joins the two solutions. It lets ions drift across to cancel the charge build-up (anions to the anode side, cations to the cathode side) — without it, both half-cells would neutralise themselves electrically and the current would stop within seconds.
- **Cell notation, one line:** anode | anode solution || cathode solution | cathode. The Daniell cell reads:

  $$\mathrm{Zn}(s) \mid \mathrm{Zn}^{2+}(1\ \mathrm{M}) \parallel \mathrm{Cu}^{2+}(1\ \mathrm{M}) \mid \mathrm{Cu}(s)$$

  The double bar marks the salt bridge; concentrations in brackets say which conditions apply.

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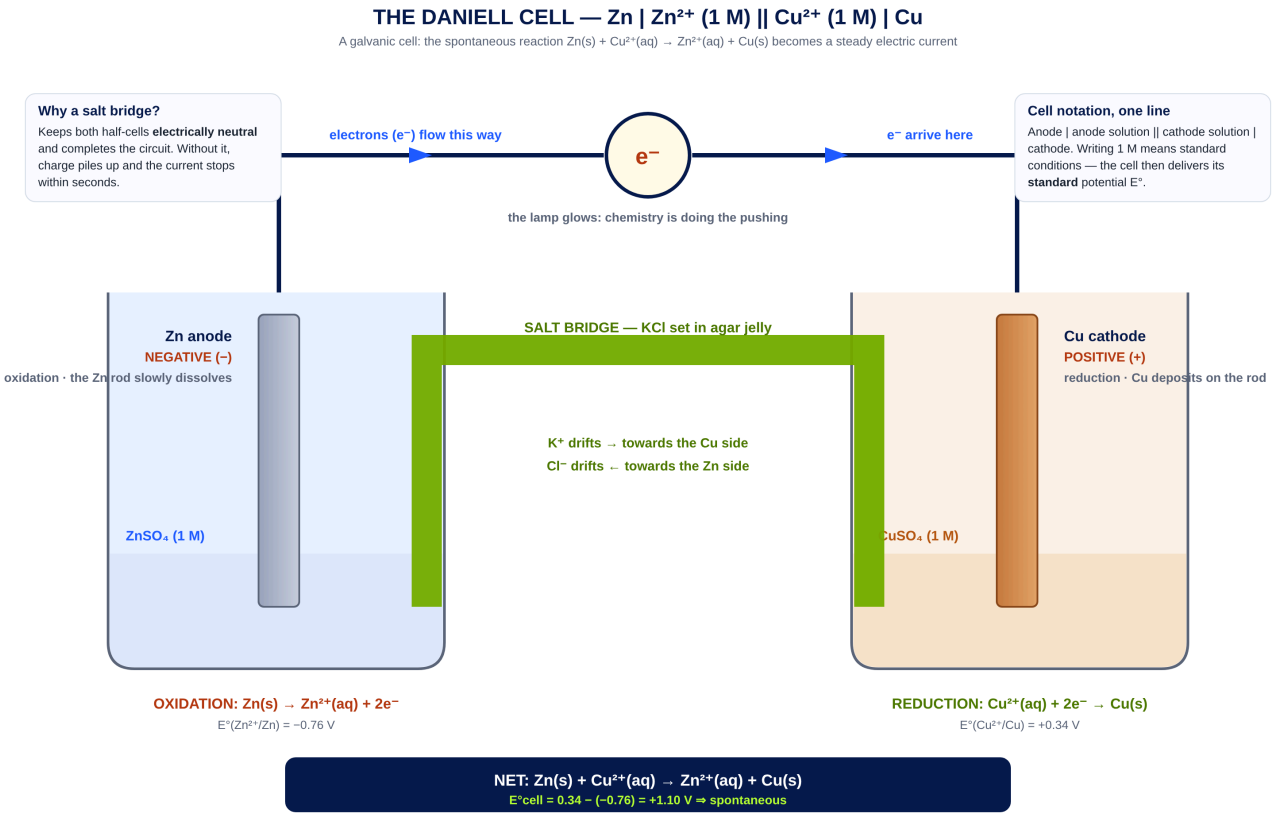


Figure 1: The Daniell cell: electron flow through the wire, ion drift through the salt bridge, and the two half-reactions with their standard potentials.

ANALOGY — THE MOUSE TRAP WATERFALL

Think of the spontaneous redox reaction as water about to fall down a cliff. In the ordinary beaker, the water simply pours down — energy released, nothing captured. A galvanic cell is a **water wheel** set in the falling stream: the fall still happens (chemistry never changes), but now the wheel turns as the water descends, and the turning wheel is your electric current. The mapping is honest: the cliff height = the potential difference (voltage), the amount of water = the quantity of charge, and the wheel's load = whatever your wire is powering. No fall, no current — which is why a dead battery is simply a reaction that has finished falling.

Standard Electrode Potential and EMF

How do we put a number on one electrode's hunger for electrons? We cannot measure a half-reaction alone — every measurement needs a complete circuit. The trick is to compare every electrode against one agreed zero point: the **standard hydrogen electrode (SHE)**.

- **The SHE:** a platinum electrode coated with platinum black, dipped in 1 M acid, with hydrogen gas bubbling over it at 1 bar and 298 K. By international agreement its potential is exactly zero:

$$\mathrm{Pt} \mid \mathrm{H_2}(1\ \mathrm{bar}) \mid \mathrm{H^+}(1\ \mathrm{M}), \quad E^\circ = 0.00\ \mathrm{V}$$

- **Measuring an electrode:** couple the electrode (as a half-cell at standard conditions — 1 M solution, 1 bar gas, 298 K) with the SHE, read the cell voltage with a voltmeter, and that reading **is** the electrode's **standard electrode potential** E°.
- **Sign convention:** E° values are written as **reduction** potentials. A positive E° means the species gains electrons more eagerly than hydrogen (a good oxidising agent); a negative E° means it is pushed into oxidation instead.
- **Landmark values to memorise:**

$$E^\circ(\mathrm{Zn^{2+}/Zn}) = -0.76\ \mathrm{V}, \quad E^\circ(\mathrm{Cu^{2+}/Cu}) = +0.34\ \mathrm{V}, \quad E^\circ(\mathrm{Ag^{+}/Ag}) = +0.80\ \mathrm{V}$$

EMF of the cell: subtract, never add

The **electromotive force (EMF)** of a cell — written E°_{cell} for standard conditions — is the potential difference between its two electrodes when no current is being drawn. With both electrode potentials tabulated as reduction potentials, one rule does all the work:

THE EMF RULE

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

Both values are **reduction** potentials from the table; you subtract the smaller (anode) from the larger (cathode). Never add two electrode potentials — that classic slip ruins every such question.

Worked line for the Daniell cell:

$$E^\circ_{\text{cell}} = 0.34 - (-0.76) = +1.10\ \mathrm{V}$$

Positive EMF → the cell reaction is spontaneous, and the cell can deliver a current. A negative EMF would mean the reverse reaction is the spontaneous one.

- **The electrochemical series:** listing electrodes by E° from most negative to most positive builds a "pushing power" ladder. Fluorine sits at the top (strongest oxidising agent), alkali metals at the bottom (strongest reducing agents).
- **The displacement shortcut:** a metal **lower** in the series displaces the ion of a metal **higher** up. Zinc (-0.76 V) displaces copper ($+0.34$ V); copper cannot displace zinc. One glance at the series settles every "which reacts with which" question.

WORKED EXAMPLE — WILL THIS CELL RUN?

A student couples an

$$\mathrm{Ag^{+}/Ag}$$

half-cell with a

$$\mathrm{Ni^{2+}/Ni}$$

half-cell. $E^\circ(\mathrm{Ag^{+}/Ag}) = +0.80$ V, $E^\circ(\mathrm{Ni^{2+}/Ni}) = -0.25$ V. Which metal is the anode, and what is E°_{cell}?

Step 1 — spot the bigger reduction potential. Silver's $+0.80$ V beats nickel's -0.25 V, so silver ions get reduced and Ag is the cathode.

Step 2 — the anode. Nickel must be oxidised: Ni is the anode, negative terminal.

Step 3 — apply the rule.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.80 - (-0.25) = +1.05 \text{ V}$$

Step 4 — interpret. Positive EMF → spontaneous. The net reaction writes itself:

$$\text{Ni}(s) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{Ag}(s)$$

— nickel metal dissolving as silver plates out.

The Nernst Equation: Potential Away from Standard Conditions

Batteries do not live at 1 M forever. As a cell runs, reactants thin out and products pile up — and the voltage **sags**. The **Nernst equation** tells you exactly how much, for any set of concentrations. For a general cell reaction with n electrons exchanged and reaction quotient Q:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

At the standard temperature of 298 K, converting to base-10 logs gives the exam-friendly form:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \left(\frac{0.059}{n}\right) \log Q$$

- **Reading Q:** products over reactants, each raised to its coefficient — exactly as in equilibrium work, with solids and pure liquids left out.
- **For the Daniell cell:**

  $$Q = \frac{[[\text{Zn}^{2+}]]}{[[\text{Cu}^{2+}]]}$$

  , so

  $$E_{\text{cell}} = 1.10 - \left(\frac{0.059}{2}\right) \log \left(\frac{[[\text{Zn}^{2+}]]}{[[\text{Cu}^{2+}]]}\right)$$

- **Concentration does the steering:** raise the product side and $\log Q$ grows, so E falls; flood the reactant side and E climbs. The cell voltage is literally a read-out of where the reaction stands.
- **Electrode form:** for a single electrode,

  $$E = E^\circ - \left(\frac{0.059}{n}\right) \log \left(\frac{1}{[[\text{M}^{n+}]]}\right)$$

  for the couple

  $$\text{M}^{n+} + ne^- \rightarrow \text{M}$$

  .

Where the equation pays off

- **Cell voltage at any moment:** plug in the concentrations, get the real EMF. This is how a battery's state of charge is inferred from its voltage.
- **Equilibrium constant from E°:** at equilibrium the cell stops pushing — $E_{\text{cell}} = 0$ and Q has become K. Substitute into the Nernst equation:

$$0 = E^\circ_{\text{cell}} - \left(\frac{0.059}{n}\right)\log K \qquad \rightarrow \qquad \log K = \frac{nE^\circ_{\text{cell}}}{0.059}$$

  A bigger E° means a bigger K — a cell reaction that nearly goes to completion.
- **pH meters in one idea:** a hydrogen electrode's potential moves by 0.059 V for every tenfold change in $[H^+]$. Measuring that drift measures pH.

WORKED EXAMPLE — NERNST AND THE EQUILIBRIUM CONSTANT

For the Daniell cell ($E^\circ_{\text{cell}} = 1.10$ V, $n = 2$): (a) find E_{cell} when $[Zn^{2+}] = 1.0$ M and $[Cu^{2+}] = 0.01$ M; (b) find K at 298 K.

Part (a) — build Q.

$$Q = \frac{1.0}{0.01} = 100$$

Apply Nernst.

$$E_{\text{cell}} = 1.10 - \left(\frac{0.059}{2}\right)\log 100 = 1.10 - (0.0295)(2) = 1.10 - 0.059 = 1.041 \text{ V}$$

The copper ions are ten times scarcer than standard, so the cell pushes 0.059 V less. (With $[Cu^{2+}] = 0.1$ M, the drop would be only half as much — the log takes care of it.)

Part (b) — equilibrium.

$$\log K = \frac{nE^\circ}{0.059} = \frac{2 \times 1.10}{0.059} = 37.3$$

So

$$K = 10^{37.3} \approx 2 \times 10^{37}$$

— an astronomically large number. Zinc displaces copper **completely**; by the time equilibrium arrives, essentially every copper ion is gone.

Gibbs Energy: The Price Tag on Spontaneity

Cell voltage measures **push**; Gibbs energy measures **useful work**. Thermodynamics ties them together, and the tie is exact — the maximum electrical work a cell can deliver equals the drop in Gibbs energy of its reaction.

THE VOLTAGE-TO-ENERGY BRIDGE

$$\Delta G = -nFE_{\text{cell}} \qquad \Delta G^\circ = -nFE^\circ_{\text{cell}}$$

Letters: n = moles of electrons transferred in the balanced cell reaction, F = **Faraday constant** = 96,500 C mol^{-1} (the charge on one mole of electrons), E in volts. The units multiply cleanly: coulomb × volt = joule.

- **Signs that mean something:** a spontaneous cell has $E^\circ > 0$ and therefore $\Delta G^\circ < 0$. Every "is this spontaneous?" question can be answered from either sign — they always agree.
- **Comparing batteries fairly:** E° alone does not rank batteries, because ΔG° also counts n. A 1.5 V cell transferring 1 electron per mole moves less energy than a 1.5 V cell transferring 2 — ΔG° catches what voltage alone hides.
- **The three-way link:**

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} = -2.303RT \log K$$

— Gibbs energy, cell voltage and equilibrium constant are three languages describing the same spontaneity. Given any one, the other two follow.

WORKED EXAMPLE — GIBBS ENERGY OF THE DANIELL CELL

Find ΔG° for the Daniell cell reaction ($n = 2$, $E^\circ_{\text{cell}} = 1.10$ V).

Apply the bridge.

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} = -(2)(96500)(1.10)$$

Arithmetic.

$$2 \times 96500 = 193000$$

, then

$$193000 \times 1.10 = 212300 \text{ J}$$

State it.

$$\Delta G^\circ \approx -212.3 \text{ kJ mol}^{-1}$$

— the reaction can, in principle, deliver 212 kJ of useful electrical work per mole of zinc dissolved. A real battery delivers less, because internal resistance wastes some of it as heat.

Conductance of Electrolytic Solutions

Metals conduct by electrons; solutions conduct by **ions** that physically crawl through the liquid. That single difference explains every oddity in this section. When a voltage is applied across two electrodes dipped in an electrolyte, ions drift — cations towards the cathode, anions towards the anode — and this ion traffic **is** the current.

- **Conductance (G):** the ease with which the solution lets current pass — the reciprocal of resistance ($G = \frac{1}{R}$, unit: siemens, S, formerly "ohm^{-1}" or mho).
- **Conductivity (κ):** the conductance of a **specific block** of solution — one centimetre cube of it, between electrodes of 1 cm² area, 1 cm apart. Unit: S cm^{-1} (or S m^{-1}). It is also called **specific conductance**.

- **The measurement problem:** you cannot clip a wire onto a solution and read κ directly — the geometry of your cell (electrode area, gap) always tags along. So we measure the **resistance** of the solution column between two platinised platinum electrodes (using an AC Wheatstone bridge — AC, because DC would electrolyse the sample) and then calibrate the cell's **cell constant** l/A with a standard KCl solution:

$$\kappa = G \times \text{cell constant} \qquad \text{cell constant} = \frac{l}{A}$$

- **Molar conductivity (Λ_m):** conductivity per mole of electrolyte — how much conduction one whole mole of dissolved electrolyte delivers:

$$\Lambda_m = \frac{\kappa \times 1000}{c}$$

with κ in S cm^{-1} and concentration c in mol L^{-1}; the unit of Λ_m is S cm^2 mol^{-1}. The factor 1000 reconciles mL (per litre of solution) with cm^3.

The dilution paradox: kappa falls, Lambda m rises

Dilute a solution and two things happen at once — and they pull in opposite directions. Keep the two units straight and the paradox dissolves.

- **κ falls on dilution:** the same litre now holds **fewer ions per cm^3**. Fewer charge carriers per unit volume means less conduction through any fixed cube. The density of the ion traffic drops.
- **Λ_m rises on dilution:** here you follow one fixed **mole** of electrolyte. As the solution thins, the ions it carries stand further apart, so their mutual attraction relaxes and each ion moves more freely. One mole of electrolyte delivers **more** conduction when given room.
- **Strong electrolytes (NaCl, KCl, HCl):** fully ionised already, so dilution gives only a gentle rise — a small boost in ion mobility. Empirically:

$$\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$$

where Λ_m° is the **limiting molar conductivity** (the value as $c \to 0$) and A is a constant for the pair of ions. Extrapolating the straight line to $c = 0$ reads off Λ_m°.
- **Weak electrolytes (acetic acid, ammonia):** only a small fraction is ionised. Dilution pushes the ionisation equilibrium **forward** (more moles ionised per mole dissolved), so Λ_m shoots up steeply — the curve never becomes straight, and you **cannot** extrapolate it to get Λ_m°. Enter Kohlrausch.

On dilution...	Strong electrolyte	Weak electrolyte
Conductivity κ	Falls (fewer ions per cm^3)	Falls (fewer ions per cm^3)
Molar conductivity Λ_m	Rises gently, almost straight	Rises sharply (more ionisation)
Λ_m° obtained by	Linear extrapolation to $c = 0$	Kohlrausch's law only — no direct extrapolation
Degree of ionisation	Already 100%	$\alpha = \Lambda_m/\Lambda_m^\circ$

Kohlrausch's Law of Independent Migration

At infinite dilution, ions are so far apart that each ion conducts completely independently of its partner — the cation and anion of an electrolyte each contribute their own fixed share, no matter whom they travelled with. That is **Kohlrausch's law**:

$$\Lambda_m^\circ = \nu_+ \Lambda_+^\circ + \nu_- \Lambda_-^\circ$$

The limiting molar conductivity of an electrolyte equals the sum of its ions' limiting molar conductivities, each multiplied by the number of that ion in the formula unit (ν_+, ν_-). Sodium chloride and sodium acetate share the same

$$\mathrm{Na^+}$$

contribution — that sharing is what makes the law powerful.

- **Reaching the unreachable:** want Λ_m° of acetic acid (a weak electrolyte you cannot extrapolate)? Subtract the strong-electrolyte pairs you **can** measure:

  $$\Lambda^\circ_{\mathrm{CH_3\,COOH}} = \Lambda^\circ_{\mathrm{CH_3\,COONa}} + \Lambda^\circ_{\mathrm{HCl}} - \Lambda^\circ_{\mathrm{NaCl}}$$

  The Na^+ terms cancel, the Cl^- terms cancel, and only

  $$\mathrm{CH_3COO^-} + H^+$$

  remains — exactly acetic acid.
- **Degree of dissociation:** for a weak electrolyte at concentration c:

  $$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

  The fraction of the molar conductivity actually delivered tells you the fraction of molecules that split.
- **Ionisation constant:** with α known, the acid dissociation constant of a weak acid HA follows from the usual equilibrium expression:

  $$K_a = \frac{c\alpha^2}{1-\alpha} \approx c\alpha^2 \quad (\text{very weak acids}, \alpha \ll 1)$$

WORKED EXAMPLE — ALPHA AND K SUBSCRIPT A OF ACETIC ACID

At 298 K, the molar conductivity of 0.05 M acetic acid is 7.4 S cm² mol⁻¹, and Λ_m° for acetic acid (assembled by Kohlrausch) is 390.5 S cm² mol⁻¹. Find α and K_a.

Step 1 — degree of dissociation.

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{7.4}{390.5} \approx 0.019$$

— under 2% of the molecules are ionised.

Step 2 — apply the ionisation-constant formula.

$$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{0.05 \times (0.019)^2}{1 - 0.019}$$

Step 3 — line by line.

$$(0.019)^2 = 3.61 \times 10^{-4}$$

, so the numerator is

$$0.05 \times 3.61 \times 10^{-4} = 1.805 \times 10^{-5}$$

; the denominator is 0.981.

Step 4 — divide.

$$K_a \approx 1.8 \times 10^{-5}$$

— the textbook value for acetic acid. The whole machinery of Λ_m measurements is really a machine for measuring acid strength without a pH meter.

Electrolytic Cells and Faraday's Laws

A galvanic cell **makes** electricity from a spontaneous reaction. Flip the purpose: pump electricity **into** a non-spontaneous reaction and force it to happen — that is **electrolysis**, and the vessel is an **electrolytic cell**. Molten salt splitting, metal electroplating, aluminium refining — all electrolysis.

- **The sign flip that sets MCQ traps:** in an electrolytic cell the **anode is positive** and the **cathode is negative** (the battery forces it so). But the **chemistry** never changes: **oxidation still happens at the anode, reduction at the cathode** — in both cell types, forever. The sign flips; the reaction assignments do not.
- **Faraday's first law:** the mass of substance deposited (or liberated) at an electrode is proportional to the charge passed:

  $$w = ZIt \qquad Z = \frac{M}{nF}$$

  Here I = current in amperes, t = time in seconds, Z = the "electrochemical equivalent" — mass liberated per coulomb. For copper ($M = 63.5$, $n = 2$),

  $$Z = \frac{63.5}{2 \times 96500} \approx 3.29 \times 10^{-4}\text{g C}^{-1}$$

  .
- **Faraday's second law:** pass the same charge through different electrolytes in series; the masses liberated are proportional to their **equivalent weights** (M/n). One coulomb does the same amount of chemical "electron bookkeeping" everywhere.
- **One mole of electrons = one Faraday = 96,500 C.** To deposit one mole of Ag ($n = 1$) needs 1 F; one mole of Cu ($n = 2$) needs 2 F; one mole of Al ($n = 3$) needs 3 F. Count the electrons in the half-reaction — that is all n ever means.

WORKED EXAMPLE — MASS DEPOSITED BY A CURRENT

A current of 1.5 A flows through a

$$\text{CuSO}_4$$

solution for 30 minutes. What mass of copper plates out? (M of Cu = 63.5 g mol^{-1}, F = 96,500 C mol^{-1}.)

Step 1 — charge passed.

$$Q = It = 1.5 \times 30 \times 60 = 2700 \text{ C}$$

Step 2 — moles of electrons.

$$n_e = \frac{Q}{F} = \frac{2700}{96500} = 0.028 \text{mol e}^-$$

Step 3 — moles of copper.

$$\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$$

, so moles of Cu = half the electron moles:

$$\frac{0.028}{2} = 0.014 \text{ mol}$$

Step 4 — mass.

$$w = 0.014 \times 63.5 \approx 0.89 \text{ g}$$

Check with the direct formula:

$$w = \frac{MIt}{nF} = \frac{63.5 \times 2700}{2 \times 96500} \approx 0.89 \text{ g}$$

— same answer, as it must be.

Predicting the products of electrolysis

What appears at each electrode depends on what is available and which species surrenders or grabs electrons most easily. Three rules sort almost every case.

- **Molten salts — only one choice.** In molten

  $$\text{NaCl}$$

  , the only cation is

  $$\text{Na}^+$$

  and the only anion is

  $$\text{Cl}^-$$

  : sodium forms at the cathode, chlorine gas at the anode. No competition, no surprises.
- **Aqueous salts — water competes.** Water itself can be oxidised to

  $$\text{O}_2$$

  or reduced to

  $$\text{H}_2$$

  . At the cathode, if the metal is **below** hydrogen's reactivity position in the series (Cu, Ag), the metal deposits; if the metal is **above** (Na, K, Ca, Mg, Al), **water** reduces to hydrogen instead. At the anode, halides (

$$Cl^-, Br^-, I^-$$

) usually win over water, but sulfate and nitrate let water win and oxygen evolves.

- **Overvoltage bends the rules:** some reactions need extra voltage to start — H_2 on certain surfaces. Classic consequence: in brine electrolysis, hydrogen still evolves at the cathode despite sodium ions being present; and industrial molten-salt electrolysis of $NaCl$ adds $CaCl_2$ to lower the melting point, since pure salt melts too hot.

Electrolyte	At the cathode	At the anode
Molten $NaCl$	Na metal	Cl_2 gas
Aqueous $NaCl$ (brine)	H_2 gas (water wins)	Cl_2 gas (halide wins) — solution turns to NaOH
Aqueous $CuSO_4$ with Cu anode	Cu deposits	Cu dissolves (anode eats away)
Aqueous $CuSO_4$ with Pt anode	Cu deposits	O_2 gas (water wins)
Aqueous $AgNO_3$	Ag deposits	O_2 gas

Batteries, Fuel Cells and Corrosion

A **battery** is a galvanic cell (or a stack of them) engineered for the real world: cheap, safe, steady. Chemistry gives two families, and the syllabus asks for exactly one example of each kind plus the fuel cell.

Primary batteries: used once, then gone

- **The dry cell (Leclanché):** the torch classic. Zinc container = anode; a carbon rod through the centre = cathode; the electrolyte is a moist paste of NH_4Cl

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and zinc chloride. The zinc shell is **consumed** as the cell runs — once the reaction inventory is spent, no recharge can rebuild it, and a punctured shell leaks.

- **The mercury cell:** zinc amalgam anode, mercury(II) oxide cathode in potassium hydroxide paste. Its speciality: the net cell reaction involves **no ions in solution whose concentration changes**, so the voltage stays essentially **constant** for its whole life — which is why it powered watches and instruments where a sagging voltage would ruin readings. (Its mercury content has retired it environmentally, but the exam idea — steady voltage — survives.)

Secondary batteries: charge, drain, repeat

- **The lead-acid (car) battery:** six cells in series, each delivering about 2 V, for roughly 12 V. Anodes of lead, cathodes of lead dioxide (

$$PbO_2$$

), electrolyte of sulphuric acid. Both electrodes convert to lead(II) sulphate during discharge:

$$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2\ O$$

(discharge) and the car's alternator pushes the same reaction backwards to recharge. The acid concentration **falls** as it discharges — a hydrometer reading of the electrolyte is literally a fuel gauge.

- **The nickel-cadmium (Ni-Cd) cell:** about 1.4 V, with NiO(OH) cathode and Cd anode in KOH. Smaller, lighter, rechargeable hundreds of times — the classic choice for portable electronics before lithium took over.

The fuel cell: a power plant that never dies

In ordinary batteries the reactants sit inside the casing; when they run out, the battery dies. A **fuel cell** feeds the reactants **from outside** continuously — so it runs as long as fuel flows in. The celebrated

$$H_2 - O_2$$

fuel cell: hydrogen is oxidised at the anode, oxygen reduced at the cathode, in hot KOH solution with porous carbon electrodes carrying a catalyst.

- **The chemistry:**

$$2H_2 + O_2 \rightarrow 2H_2\ O$$

— the only product is water, which on crewed space missions was not waste but **drinking water**.

- **The efficiency edge:** a heat engine tops out near 40 per cent efficiency; fuel cells convert about 70 per cent of the reaction energy directly into electricity — no burning, no boiler, no steam turbine.
- **Why it matters:** continuous clean power for spacecraft, backup installations and (increasingly) vehicles — the battery that refuels like a car.

Corrosion: electrochemistry you can see rusting

Corrosion is the slow eating-away of a metal by the environment — and rusting of iron is a galvanic problem, not simple "air attack". In a drop of moisture on an iron surface, uneven oxygen access creates tiny anodic and cathodic patches, and the metal itself supplies the wire.

- **At the anodic spot:**

$$\mathrm{Fe}(s) \rightarrow \mathrm{Fe}^{2+}(\mathrm{aq}) + 2e^{-}$$

  — iron dissolves into the water droplet.
- **At the cathodic spot (plenty of oxygen):**

$$\mathrm{O_2} + 2\mathrm{H_2\,O} + 4e^{-} \rightarrow 4\mathrm{OH}^{-}$$

  .
- **The rust forms second-hand:**

$$\mathrm{Fe}^{2+}$$

  ions meet

$$\mathrm{OH}^{-}$$

  and air, oxidising further to hydrated iron(III) oxide — the familiar red-brown flakes,

$$\mathrm{Fe_2O_3 \cdot}\, x\mathrm{H_2\,O}$$

  , which neither stick nor protect, so the attack continues inward.
- **Prevention, method by method: barrier coatings** (paint, grease, enamel) simply seal out air and water; **alloying** (stainless steel with chromium) builds a self-repairing oxide film; **galvanisation** coats iron with zinc — and here is the beautiful part: zinc, being **more** reactive, becomes the anode and corrodes **instead of** the iron even when the coating scratches (sacrificial protection); **cathodic protection** connects the iron to a more easily oxidised metal (magnesium blocks on pipelines and ship hulls) or forces it negative electrically, so the iron can only ever be the cathode — and cathodes do not corrode.

ANALOGY — THE BODYGUARD RULE

Sacrificial protection works like a bodyguard who takes every hit. Zinc and iron walk together; when trouble (oxidation) shows up, zinc — the more willing fighter — absorbs it and "dies" first, while the protected iron stays untouched. The mapping is electrochemically exact: the bodyguard must be **more** reactive (lower in electrode potential) so that **he** becomes the anode. Paint, by contrast, is just a locked door — protective only until it cracks, which is why the bodyguard strategy beats the locked door wherever scratches are guaranteed (ships, pipelines, fences).

Key Reactions & Rules: One-Stop Revision

CELL VOLTAGE, NERNST, CONDUCTIVITY, FARADAY

Rule / formula	Symbols in plain words	When to reach for it
$E^\circ_{\text{cell}} = E^\circ_{\text{cat}} - E^\circ_{\text{an}}$	both are reduction potentials — subtract, never add	Every EMF question, ever
$E = E^\circ - \left(\frac{0.059}{n}\right)\log Q$	Q = products/reactants at 298 K	Cell voltage at non-standard concentrations
$\log K = \frac{nE^\circ_{\text{cell}}}{0.059}$	valid because E = 0 at equilibrium	Converting voltage to equilibrium constant
$\Delta G^\circ = -nFE^\circ_{\text{cell}}$	F = 96,500 C mol^{-1}; coulomb × volt = joule	Voltage to Gibbs energy and back
Pt | H_2(1 bar) | H^+(1 M)	SHE, the zero reference	Defining standard potentials
$\kappa = G \times \left(\frac{l}{A}\right)$	cell constant $\frac{l}{A}$ from KCl calibration	Conductivity measurements
$\Lambda_m = \frac{\kappa \times 1000}{c}$	c in mol L^{-1}; unit S cm^2 mol^{-1}	Molar conductivity from conductivity
$\Lambda_m = \Lambda^\circ_m - A\sqrt{c}$	strong electrolytes only	Extrapolating to infinite dilution
$\Lambda^\circ_m = \nu_+\Lambda^\circ_+ + \nu_-\Lambda^\circ_-$	Kohlrausch: ions migrate independently	Weak electrolytes' Λ°_m by subtraction
$\alpha = \frac{\Lambda_m}{\Lambda^\circ_m}$	fraction ionised at concentration c	Weak acid strength problems
$K_a = \frac{c\alpha^2}{1-\alpha}$	$\approx c\alpha^2$ when α is tiny	From conductivity to acid strength
$w = \frac{MIt}{nF}$	I in amperes, t in seconds	Every Faraday's-law mass question
1 F = 96,500 C mol^{-1}	charge of one mole of electrons	Converting charge to electron moles
$E^\circ < 0 \rightarrow \Delta G^\circ > 0$	non-spontaneous — needs electrolysis	Deciding galvanic vs electrolytic

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
Anode is always negative	In an **electrolytic** cell the anode is positive	Oxidation happens at the anode in BOTH cell types — only the sign flips
Add the two electrode potentials	You must subtract: $E^\circ_{\text{cat}} - E^\circ_{\text{an}}$	An answer built by adding the **signed** values ($0.34 + (-0.76) = -0.42$ V) is the classic bait
Dilution lowers everything	κ falls but Λ_m **rises** on dilution	A "what happens to molar conductivity on dilution" check — the answer is rise, for both strong and weak
Weak electrolytes can be extrapolated	Their curve never straightens — use Kohlrausch subtraction	Reading Λ°_m of acetic acid from its own dilution curve is impossible by definition
Sodium appears first in brine electrolysis	Water reduces before Na^+ can	Brine gives H_2 at the cathode and NaOH in solution — not sodium metal
Chlorine always wins at the anode	Sulfate/nitrate let water through: O_2 evolves	Match the anion: halides vs oxyanions give different gases
Battery voltage measures its energy	Voltage alone ignores n — energy is $\Delta G^\circ = -nFE^\circ$	Two 1.5 V cells can store different energies if n differs
Rust is just iron + oxygen	Water and ion traffic are essential — dry iron does not rust	The mechanism is a miniature galvanic cell inside a moisture droplet

The Whole Chapter in One Picture

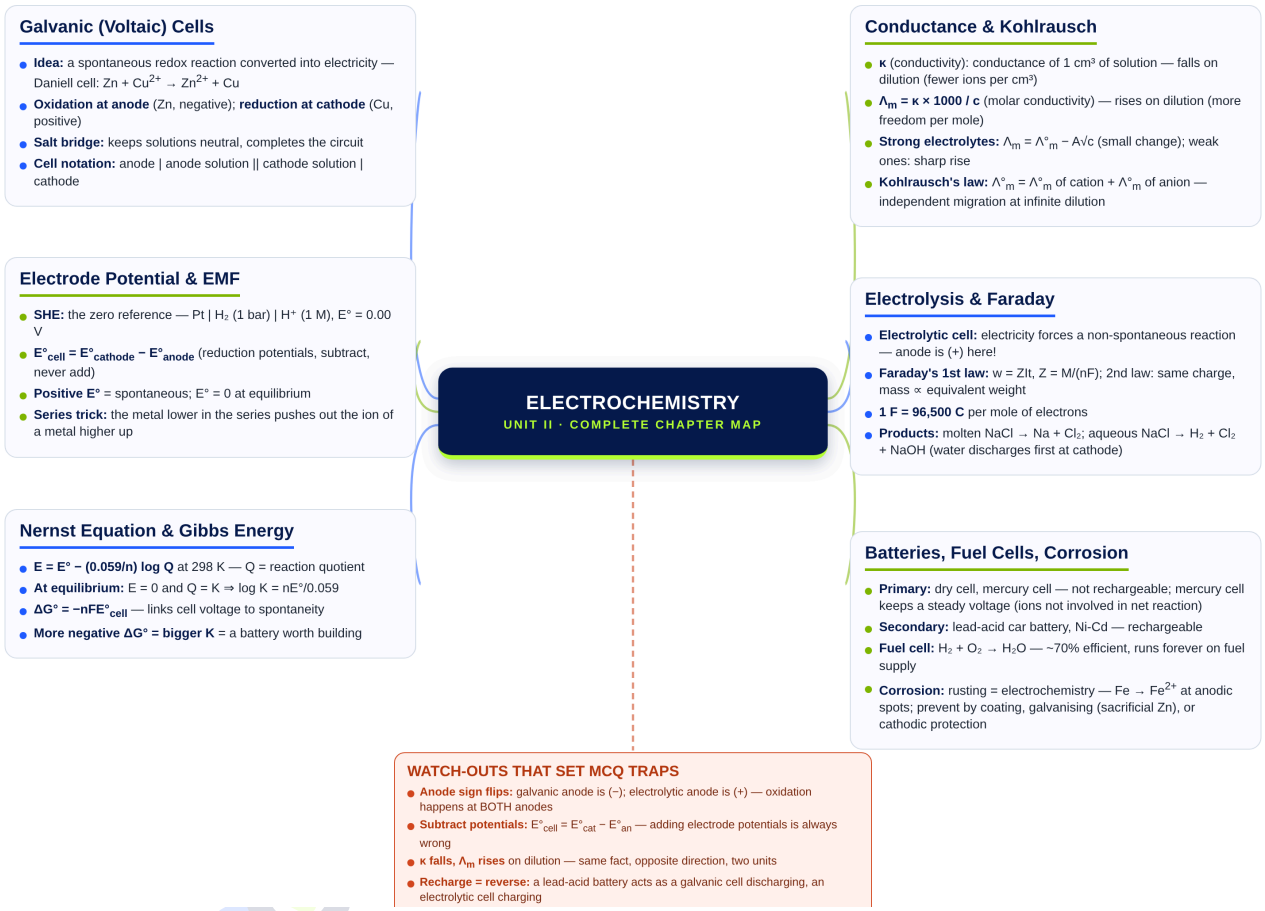


Figure 2: Chapter mindmap: galvanic cells, electrode potentials and EMF, the Nernst equation and Gibbs energy, conductance and Kohlrausch's law, electrolysis with Faraday's laws, and batteries, fuel cells and corrosion — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and −1 for every incorrect answer.

Q1. In a galvanic cell, oxidation takes place at the electrode that is:

(A) negative and called the cathode
(B) positive and called the anode
(C) negative and called the anode
(D) positive and called the cathode

Q2. For the cell Zn | Zn^{2+} || Cu^{2+} | Cu with $E^{\circ} = -0.76$ V for zinc and $+0.34$ V for copper, the standard EMF is:

(A) $+0.42$ V
(B) $+1.10$ V
(C) -1.10 V
(D) $+0.76$ V

Q3. The standard hydrogen electrode is assigned a potential of:

(A) +1.00 V **(B)** 0.00 V
(C) −1.00 V **(D)** 0.76 V

Q4. As the concentration of

$$Cu^{2+}$$

in a Daniell cell is decreased, the cell voltage will:

(A) increase **(B)** decrease
(C) stay exactly the same **(D)** first rise, then fall

Q5. The molar conductivity of a strong electrolyte on dilution:

(A) falls sharply because ions disappear **(B)** rises gently towards a limiting value
(C) remains constant **(D)** first falls, then rises

Q6. The limiting molar conductivity of a weak electrolyte like acetic acid is best obtained by:

(A) extrapolating its own dilution curve **(B)** Kohlrausch's law using strong electrolytes
(C) multiplying its conductivity by 1000 **(D)** measuring it at high concentration

Q7. In the electrolysis of aqueous NaCl (brine), the product at the cathode is:

(A) sodium metal **(B)** hydrogen gas
(C) oxygen gas **(D)** sodium hydroxide

Q8. A current of 2 A deposits copper from

$$CuSO_4$$

solution. The charge needed to deposit one mole of Cu is about:

(A) 96,500 C **(B)** 48,250 C
(C) 193,000 C **(D)** 24,125 C

Q9. In a lead-acid battery being **discharged**:

(A) both electrodes turn into lead sulphate **(B)** sulphuric acid concentration rises
(C) lead dioxide is produced at the anode **(D)** the cell behaves as an electrolytic cell

Q10. Galvanised iron stays protected even when the zinc coating is scratched, because zinc:

(A) blocks air more completely than paint **(B)** is less reactive than iron
(C) acts as a sacrificial anode and corrodes first **(D)** coats the scratch with oxide instantly

OurMocks

Detailed Solutions

Sol. Q1. The chemistry is fixed: oxidation always happens at the **anode**. In a galvanic cell the anode releases electrons into the wire, so it is the **negative** terminal. Hence: negative, and called the anode. The sign flips only in electrolytic cells.

Answer: (C)

Sol. Q2. Apply the EMF rule: $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 - (-0.76) = +1.10$ V. Copper is reduced (bigger potential), zinc is oxidised. Watch the sign logic: the anode potential enters **with its own negative sign**, so adding the signed values $(0.34 + (-0.76) = -0.42$ V) is the classic wrong answer.

Answer: (B)

Sol. Q3. By international agreement the standard hydrogen electrode (Pt, 1 bar

$$\text{H}_2$$

, 1 M H^+) is the zero of the potential scale — every other electrode is measured **against** it. Zinc's −0.76 V literally means "0.76 V below hydrogen".

Answer: (B)

Sol. Q4. Nernst: $E_{\text{cell}} = 1.10 - \left(\frac{0.059}{2}\right)\log\left(\frac{[[\text{Zn}^{2+}]]}{[[\text{Cu}^{2+}]]}\right)$. Lowering $[\text{Cu}^{2+}]$ makes the fraction bigger, its log bigger, and the subtracted term bigger — so E **falls**. Scarcer reactant, weaker push.

Answer: (B)

Sol. Q5. Strong electrolytes are fully ionised already; dilution only relaxes ion-ion attractions, so Λ_m rises gently along $\Lambda_m = \Lambda^\circ_m - A\sqrt{c}$ towards the limiting value. Conductivity κ falls — but the question asks about molar conductivity.

Answer: (B)

Sol. Q6. The acetic acid curve rises too steeply and never straightens, so extrapolation is impossible. Kohlrausch's law assembles Λ°_m from measurable strong electrolytes: $\Lambda^\circ_{\text{CH}_3\text{ COOH}} = \Lambda^\circ_{\text{CH}_3\text{ COONa}} + \Lambda^\circ_{\text{HCl}} - \Lambda^\circ_{\text{NaCl}}$.

Answer: (B)

Sol. Q7. Water is reduced more easily than

$$\text{Na}^+$$

in aqueous solution, so the cathode makes

$$\text{H}_2$$

gas (and the solution near it turns to NaOH). Sodium metal is only obtainable from **molten** NaCl.

Answer: (B)

Sol. Q8. Depositing Cu needs 2 electrons per atom, so 1 mol Cu needs 2 mol electrons: $Q = 2 \times 96500 = 193000$ C. One mole of electrons (96,500 C) would deposit only half a mole.

Answer: (C)

Sol. Q9. On discharge, lead and lead dioxide both convert to lead(II) sulphate — that single fact is the discharge equation in one line. The acid is consumed (concentration **falls**), and a discharging lead-acid battery is a **galvanic** cell, not electrolytic.

Answer: (A)

Sol. Q10. Zinc sits **below** iron in the electrochemical series (more easily oxidised). At a scratch, zinc becomes the anode and dissolves instead of the iron — sacrificial protection. Paint can only block; zinc protects electrically.

Answer: (C)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q2 built by adding	Treating EMF as a sum	Subtract: $E^\circ_{\text{cell}} = E^\circ_{\text{cat}} - E^\circ_{\text{an}}$ — the anode value keeps its negative sign
Q5 option A	Confusing κ with Λ_m	κ falls on dilution; Λ_m rises — the question named molar conductivity
Q7 option A	Forgetting water's competition	Water wins at the cathode; sodium needs molten salt
Q8 option A	Counting electrons per mole wrong	Read the half-reaction: Cu needs 2 e^- each, so one mole costs two Faradays
Q9 option C	Mixing discharge with charging	Discharge consumes PbO_2 and acid; charging reverses it

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